

Secure Authentication and Attack Detection System

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1. Problem Statement / Project Idea

Authentication systems are a primary target for cyberattacks. Many applications remain vulnerable due to improper password storage, lack of input validation, absence of attack detection mechanisms, and weak monitoring systems. Common attacks such as SQL Injection and Brute-Force Login Attempts exploit insecure authentication implementations, leading to unauthorized access and data breaches.

The purpose of this project is to design and implement a Secure Authentication and Attack Detection System that demonstrates both authentication vulnerabilities and their corresponding defense mechanisms. The system will implement secure password storage using hashing and salting techniques, detect SQL injection patterns in login inputs, prevent brute-force attacks through threshold-based monitoring, enforce account lockout policies, and maintain structured audit logs for security analysis.

The objective is to build a security-focused authentication engine that emphasizes implementation of real-world security controls rather than user interface or database complexity. The project will simulate attack scenarios and demonstrate how properly implemented defense mechanisms mitigate these threats.

2. Scope

The project will implement the following security-focused modules:

- **Secure User Registration**

- Password hashing using SHA-256
- Random salt generation
- Secure storage of salted hashes in database
- **Secure Login Mechanism**
 - Password verification through salted hash comparison
 - Login attempt tracking
 - Automatic account lockout after multiple failed attempts
 - Time-based unlock mechanism
- **SQL Injection Detection Module**
 - Identification of suspicious input patterns
 - Logging of malicious attempts
 - Blocking of detected injection attempts
- **Brute-Force Detection System**
 - Monitoring repeated failed login attempts
 - Time-window based threshold analysis
 - Temporary account blocking
- **Audit Logging System**
 - Recording every authentication attempt
 - Timestamp and simulated IP logging
 - Status tracking (Success / Failed / Blocked)
- **Administrative Module (Console-Based)**
 - View and analyze authentication logs
 - Unlock blocked accounts
 - Review attack statistics

Limitations / Boundaries

- The system will be implemented as a console-based application; advanced UI development is not included.
- Network-level packet analysis is outside the scope; attack scenarios will be simulated within the application.
- The project focuses specifically on authentication security and attack detection, not on full enterprise-level deployment.
- Machine learning techniques are not included, as the focus is on core security mechanisms.

This project strictly aligns with Information Security principles by implementing authentication security, vulnerability demonstration, attack detection, and defensive controls within a structured and practical system architecture.